Predicting the efficacy of malaria vaccines across the life-course

Supplementary Notebook 1. Overview of the Edinburgh model

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1 Introduction

Current state-of-the-art epidemiological models of malaria and its interventions are complex, individual-based models with large numbers of parameters that require a deep knowledge of mathematical modelling and expert computational skills to use (Galactionova, Smith, and Penny 2021). Our motivation is to develop a small, deterministic, population-based model with a minimal number of parameters, all estimated from empirical data, that explain age patterns of clinical malaria, severe malaria and deaths. We also require that the model provide fast simulations for responsive visualisations for a public website. The website is designed to allow non-modelling scientists to explore the sensitivity of age patterns to changes in parameter values and different intervention strategies.

In this notebook we summarise the main aspects of our model. In the accompanying notebooks we describe the development and justification of the model in more detail.

As with all malaria epidemiological models, ours is a simplification of the highly complex processes underlying malaria transmission, disease and control. However, such models clarify thinking about relationships between malaria immunology and epidemiology and vaccine characteristics and their implementation. They help anticipate and understand counter-intuitive effects in particular endemic settings (Halloran et al. 1989). And they allow us to ask, What if? questions. Therefore, numerical results should not be interpreted as precise predictions, but rather as providing qualitative insights for informing vaccine development, clinical trial design and implementation of vaccine programmes.

2 Demography

The model follows a single multi-age cohort of people weekly from birth till last death. This is in contrast to other malaria intervention modelling studies which follow a multi-cohort population of individuals maintained at constant size by births and deaths (see Notebook S2 for our justification for taking this approach).

All-cause, age-specific death rates in the cohort are calculated from the Tanzania 2019 life table (Figure 1). As most severe malaria episodes and malaria-associated deaths occur in the under fives, modelling all-cause mortality till old age is not really necessary. We include it to avoid having to set an arbitrary upper age limit and to accurately calculate lifetime cumulative clinical and severe episodes and deaths.

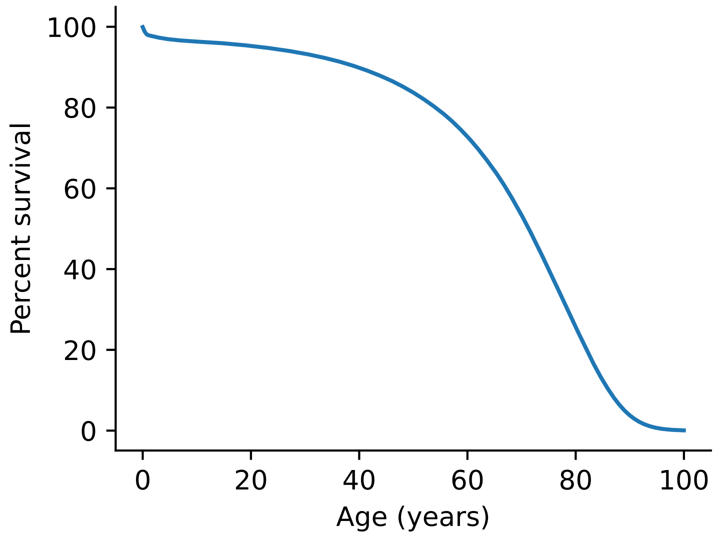

Figure 1: Survival function of cohort based on the 2019 Tanzania life table.

3 Blood infection rate as a measure of transmission intensity

The field studies that we use to parameterise our models of clinical and severe malaria immunity intensively monitored their populations with active and passive surveillance (Trape and Rogier 1996; Gonçalves et al. 2014; Trape et al. 2024). Clinical attacks were immediately drug treated, which means that most were likely to be from independent mosquito bites rather than recrudescences of earlier infections.

Our models of the development of clinical and severe malaria immunity are based on exposure, defined as the cumulative number of blood infections from independent mosquito bites over time. In our model, immunity does not depend on the cumulative parasite load (Maire et al. 2006a) or the number of antigenically diverse parasites experienced (Eckhoff 2012). We also assume that presentation of clinical or severe symptoms are immediately drug treated and all parasites cleared. Therefore in our model there can be only one clinical attack per independent blood infection, i.e., there is no recrudescence. In addition, we assume that any infection that does not result in clinical symptoms is naturally cleared over some unspecified timescale and that such infections have no effect whatsoever on further infections.

Given this, it makes sense for our model to use the rate of independent blood infections as a measure of transmission intensity. In fact, we use annual average number of independent blood infections per child. We could have used EIR as a measure of transmission intensity, like other models (e.g., Penny et al. (2016)), but this introduces additional model parameters to convert from EIR to blood infection rate (Smith et al. 2004); and we are trying to minimise the number of parameters in the model.

Our rationale for using independent blood-stage infections as a measure of transmission intensity is based on biological plausibility, empirical evidence, and modelling considerations, as discussed below.

3.1 Biological plausibility: clinical immunity is primarily disease-modifying, not infection-blocking

Epidemiological evidence indicates that individuals in endemic settings often acquire protection against clinical malaria without acquiring sterilising immunity against infection. For example, in Mali, Tran et al. (2013) showed that the risk of acquiring PCR-detectable infection did not fall with age, whereas the risk of clinical malaria did. Which implies that immunity reduces symptoms given an infection rather than preventing infection outright. In a Kenyan cohort designed to separate immunity from absence of exposure, Bejon et al. (2009) similarly observed an age-related shift from febrile malaria to asymptomatic parasitaemia without a corresponding reduction in infection incidence.

If immunity acts mainly by reducing the probability of clinical disease conditional on infection, then the natural “dose counter” for immunity building is not the cumulative number of infectious bites, but the cumulative number of blood-stage infections that successfully establish and present blood-stage antigen. This motivates cumulative blood-stage infections as a predictor for temporal immunity acquisition and blood-stage infection rate as a natural and simple measure of transmission intensity.

3.2 Empirical support: prior blood-stage infections predict reduced clinical risk upon infection

Quantitative support for using cumulative blood-stage infections comes from analyses that jointly estimate exposure heterogeneity and immunity effects. In a high-transmission Ugandan cohort with intensive follow-up, Rodriguez-Barraquer et al. (2016) estimated that the risk of developing clinical malaria upon infection declined with both age and the cumulative number of prior infections, with an average reduction of ~2% per additional prior infection and ~6% per additional year of age. (Their conclusion of an age effect is not adequately supported, though, because (i) their exposure-only model (c2) had lower DIC than their age-only model (c1) and their exposure+age model (c1a) had the same DIC as their exposure-only model (c2) and (ii) in their models adjusted for treatment, no comparison between exposure-only and age-only models was made (their Table S2).)

3.3 Interventions that reduce blood-stage infections delay clinical immunity

Chemoprophylaxis and other exposure-reducing interventions function as experiments that manipulate cumulative infections. If cumulative infections drive clinical immunity acquisition, then suppressing infections should delay immunity and produce a transient rebound in morbidity. For example, in Tanzanian infants receiving prophylaxis in the first year of life, Aponte et al. (2007) found large reductions in clinical malaria during prophylaxis, followed by rebound of clinical malaria for about two years after stopping. In The Gambia, Greenwood et al. (1995) observed increased clinical attacks in the year after stopping long-term chemoprophylaxis.

3.4 Why blood infections rather than inoculations: force-of-infection saturates relative to EIR

Several modelling studies demonstrate that the force of infection increases sub-linearly with EIR because (i) some sporozoite inoculations fail to establish infection, (ii) heterogeneity in biting concentrates exposure in a few individuals, and (iii) pre-erythrocytic immunity reduces the probability that a bite produces blood-stage infection as exposure accumulates (Smith et al. 2004; 2006). Smith et al. (2004) have argued that EIR is a poor indicator of infection rate across ages and across populations.

Because clinical immunity is driven by the experience of blood-stage infection, cumulative blood-stage infections provides a more proximal predictor than cumulative inoculations.

3.5 Consistency with mechanistic formulations of clinical immunity in transmission models

Many malaria transmission models represent clinical immunity as a function of cumulative exposure, sometimes using explicit immunity variables that increase with parasite densities or infection events and decay over time (Penny et al. 2016). Within OpenMalaria, clinical disease arises via a pyrogenic threshold that responds dynamically to parasite load, and immunity components are calibrated to reproduce age- and exposure-dependent patterns of clinical incidence (Smith et al. 2006; Maire et al. 2006b; Ross et al. 2006). While these models often use cumulative parasite density or “time infected” internally, they support the same conceptual structure: repeated blood-stage infections produce gradual reductions in clinical risk.

Empirically, Smith et al. (1998) reported that clinical incidence in very young Tanzanian children decreased steeply with estimated cumulative time infected with trophozoites, rather than with estimated cumulative inoculations since birth. Cumulative blood-stage infections is not identical to cumulative time infected, but under broadly stable infection durations and frequent reinfection, the count of independent blood-stage infections is a pragmatic proxy for cumulative antigenic and inflammatory experience in blood-stage.

Importantly, cumulative infections is also easier to define consistently across cohorts than integrals of (often imperfectly detected) parasitaemia.

3.6 Counting independent blood infection rates in the field

The problem with using independent blood infections as transmission intensity is that these are not so easily defined and counted in the field if surveillance is not active and frequent, infection is confirmed by only microscopy and not by more sensitive methods such as qPCR and presentation is not immediately followed by drug treatment clearing 100% of parasites. So translating our model outputs, which are in terms of independent blood infections, to field conditions needs care.

Having said that, our key results and conclusions concern independent blood infection rates below about 3 per year. Counting independent blood infections below this threshold should be less fraught with confounding by recrudescence and super-infection than at higher transmission intensities. That is, at low transmission intensities, estimates of blood infection rates are likely to be more accurate than at high transmission intensities. In addition, below this threshold, and in populations with comprehensive drug treatment upon clinical presentation, our modelling has shown that there is an almost linear relationship of about 1 annual clinical attack in the under fives for about every 2 annual blood infections (Notebook S11). Thus counting average annual clinical attacks in the under fives may give a reasonable indication of the average annual blood infection rate.

4 Our model of infection, disease and death

An infectious bite can lead to a liver infection which can lead to a blood-stage infection. We are not modelling the insect vector population, nor transmission between hosts. Moreover, we do not model infection blocking immunity as that develops relatively slowly and only in those with very high cumulative exposure (Maire et al. 2006a). This means we can ignore transmission and pre-erythrocytic infections and focus solely on blood-stage infections. Figure 2 illustrates our model of the time course from blood infection to potential death.

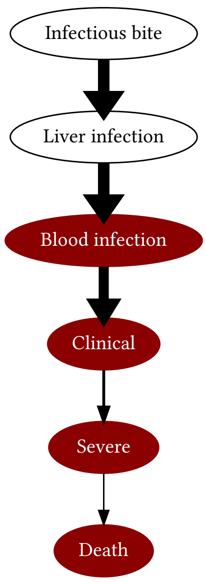

Figure 2: Time course of a malaria infection. Once established, a blood infection may resolve without disease (uncomplicated malaria) or may progress to clinical disease (i.e., fever). It may then resolve or progress to severe malaria which may result in death.

Let λ be the population-average annual blood infection rate per child. Time is discretised in to one-week intervals. In each week the model calculates the number of new infections in children of age a who have previously experienced n blood-stage infections. Blood infection rate is modified by one or more of the following:

1. seasonal changes in infection rate

2. heterogeneity in bite rate/susceptibility (these are epidemiologically equivalent in terms of modifying infection rate)
3. early life protection from, for example, maternal antibodies and reduced bite rate due to smaller body surface area relative to adults
4. protection derived from interventions such as vaccination and chemoprevention

Seasonal variation in the infection rate is modelled using the circular Von-mises distribution with configurable mean and variance to model different timings and lengths of the malaria season. See Notebook S3 for more details.

Relative bite rate/host susceptibility is assumed to have a symmetric triangular distribution centred on one. Smith et al. (2005) fitted a Gamma distribution to prevalence data. However, the unconstrained upper bound of the Gamma distribution introduces numerical instabilities. So, for pragmatic reasons, we use the constrained triangular distribution based on Figure 3A of Rodriguez-Barraquer et al. (2016).

Early in life children are at less risk of a blood infection than older children. They are protected from blood infection by maternal antibodies, and their smaller body surface area means they are less likely to be bitten by mosquitoes (see Figures 1 of Smith et al. (2004) and Smith et al. (2006)). We could model maternal antibody and body surface area protection separately, as the WHO consensus models did (Penny et al. 2016). But to keep the model as simple as possible we combine them into a single "early-life protection factor" against blood infection. This assumption is supported by our modelling of clinical episodes in two Senegal villages, Dielmo and Ndiop (Notebook S5, Trape and Rogier (1996), Trape et al. (2024)). We assume that relative early life protection decays exponentially from a maximum of ρ_{elp} at age zero and with a half-life of τ_{elp}:

$$\text{Relative protection from blood infection at age } \; a = \rho_{\text{elp}} \exp\left(-\frac{\ln 2}{\tau_{\text{elp}}} a\right)$$

This simple model fitted the data from the two villages very well with estimates $\rho_{\text{elp}} = 0.71 \pm 0.03$ and $\tau_{\text{elp}} = 1.73 \pm 0.42$ years. That is, newborns are at 71% less risk of infection than adults, and risk of infection doubles roughly every 1.7 years in childhood.

5 Disease

In our model, a blood-stage infection can either progress to febrile disease (clinical episode) or is asymptomatic and cleared. Based on our modelling of data in (Gonçalves et al. 2014) (see Notebook S4), we estimate that 21.1% ± 1.4% of blood-stage infections are asymptomatic. Griffin, Ferguson, and Ghani (2014) estimate a very similar value.

A clinical episode may be mild (also referred to as uncomplicated) and is then cleared. Or it may progress to become severe using the WHO definition of severe disease. We do not distinguish between the various syndromes of severe disease.

We assume that only one clinical episode can occur per infection. This, of course, is not always true. But Smith et al. (2004) have argued: *"There is strong evidence both from molecular typing and from patterns of seasonality in morbidity that clinical malaria is normally caused by newly invading parasites, and the most severe symptoms generally accompanying the first peak of parasite density after infection."*

6 Death

There is much qualitative and quantitative uncertainty about malaria associated deaths. Therefore, all models' predictions are subject to considerable uncertainty. Consensus modelling, in which the outputs of different models with different underlying assumptions are compared, is a standard method to mitigate against some of this uncertainty (Penny et al. 2016). But even when models agree, this method is no guarantee that the consensus output is an accurate reflection of reality if model assumptions and parametrisations are inaccurate.

It has been argued that there are two major causes of malaria-associated deaths: directly attributable deaths due to severe malaria, and indirectly attributable deaths that would be prevented in the absence of a malaria infection due to co-morbidities (Molineaux 1997; Ross et al. 2006).

Of the WHO consensus models, only OpenMalaria models both of these causes (Penny et al. 2016). We tried to implement indirect deaths using the model of Ross et al. (2006), but were unable to match the output of our model to the shape of the all-cause mortality data presented in their Figure 9. In addition, including indirect deaths does not match age-patterns observed from verbal autopsy reports (Abdullah et al. 2007). We therefore decided not to implement indirect deaths.

Age-specific, hospitalised, severe malaria case fatality rates (CFR) are taken from Reyburn et al. (2005), and shown in Figure 3. Our analysis of this dataset is detailed in Notebook S7.

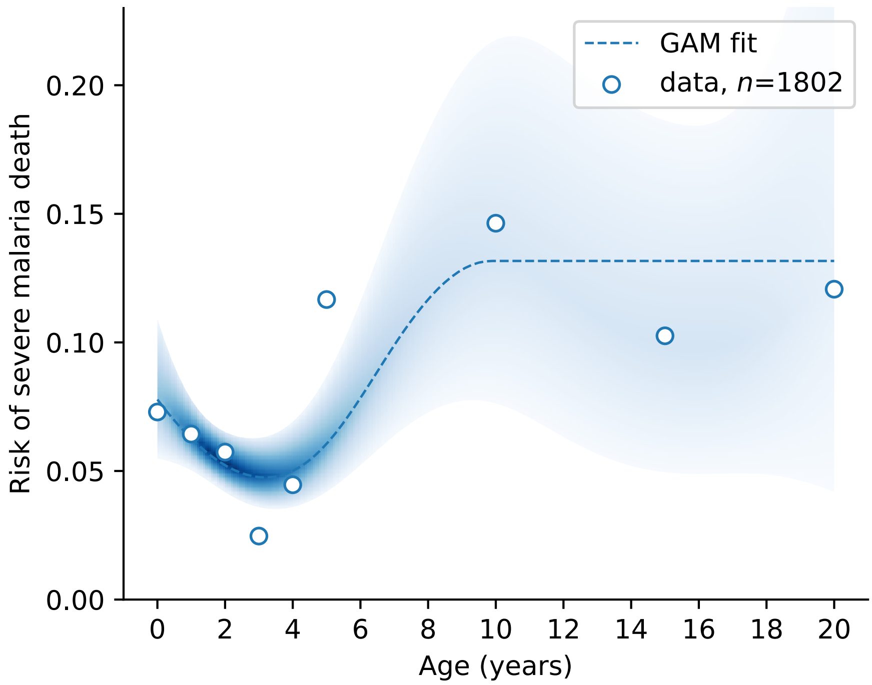

Figure 3: Age specific, hospitalised, severe malaria case fatality rates from Reyburn et al. (2005) were fitted by logistic GAM. Open circles: age-stratified CFRs calculated from the n=1802 deaths. Shaded blue band: 95% CI of the GAM-estimated age-specific CFR. Dashed blue line: the estimated mean CFR from 0-10 years old, from age 10+ we assume a constant CFR as the GAM estimate becomes unreliable due to very few adult deaths. Note, the GAM is fitted to the full set of deaths, not the age-stratified values shown as open circles.

OpenMalaria also use this data (Ross et al. 2006). The Imperial model infers age-specific CFR by using data on severe malaria syndromes. Their estimated CFR, though, is almost flat, resulting in an almost constant CFR with age (see Figure 3D in Griffin et al. (2015)). The other two WHO consensus models do not use age-specific CFRs (Penny et al. 2016).

In our model we use the estimated mean CFR shown as the dashed blue line in Figure 3.

7 Naturally acquired immunity

Levels of chemoprevention and immunity (naturally acquired or by immunisation) are represented as within-cohort averages in our model.

Figure 4 illustrates the two types of malaria immunity and their affect on the progression of infection and disease that we include in our model. We do not model pre-erythrocytic, infection blocking immunity as it develops relatively slowly and only in those with very high exposure (Maire et al. 2006b; Bejon et al. 2009; Tran et al. 2013). Anti-parasite immunity blocks merozoite invasion of erythrocytes and disrupts parasite development in erythrocytes. It limits parasite growth and peak parasitaemia. It therefore has two main effects: it reduces the risk of a blood infection initiating (which we do not model) and it reduces the risk of clinical symptoms developing (which we do model). This so called clinical immunity is assumed to develop slowly over repeated infections. Immunity to severe malaria, on the other hand, develops more rapidly. We do not model protective immunity against gametocytes as we are not modelling between-host transmission.

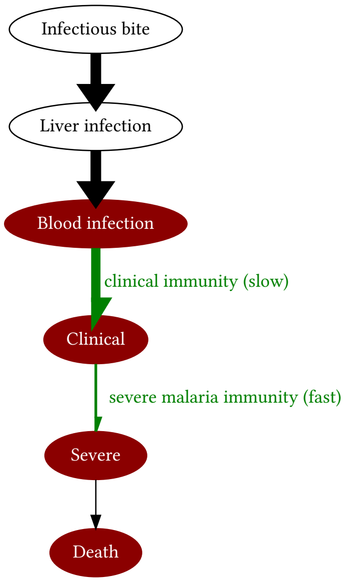

Figure 4: The modelled types of malaria immunity and where they act in the infection process

As discussed above, previous models have assumed that blood-stage infection is the main driver of naturally acquired immunity (Penny et al. 2016). They all assume that protection increases with each blood-stage infection. Some models also assume that protection increases with cumulative parasitaemia (OpenMalaria) or age (Imperial). Apart from the GSK model, the other WHO consensus models also assume that immunity can decay between infections.

We do not model immunity per se. Instead, we model the risk of clinical malaria given a blood infection and the risk of a clinical episode becoming severe. We assume that the risk of clinical malaria decays with the cumulative number of blood infections (also termed exposure). And we assume that the risk of a clinical episode becoming severe decays with age-dependent exposure. Between infections risk is assumed to remain constant, i.e., there is no decay in immunity.

7.1 Naturally acquired immunity to clinical malaria

We tested several models of the reduction in risk of clinical malaria with exposure (see Notebook S5). We used clinical episode rate data from two Senegalese villages: Dielmo and Ndiop. These populations have been uninterruptedly followed with identical and strict clinical surveillance programs. These include daily home visits to each person and the 24h presence of medical teams in the villages to diagnose and treat any pathological episode (Trape and Rogier 1996; Trape et al. 2024). Thus these data are of very high quality. We found that a sigmoidal decay with exposure gives excellent fits to individual-level data from Dielmo village and population-level data from both Dielmo and Ndiop. The form of the decay is given by the equation:

$$\Pr(\text{clinical episode on } n^{\text{th}} \text{ infection}) = \rho_{\text{clinical}} \frac{e^{\delta_{\text{clinical}}[\gamma_{\text{clinical}}-1]} - 1}{e^{\delta_{\text{clinical}}[\gamma_{\text{clinical}}-1]} - 2 + e^{\delta_{\text{clinical}}[n-1]}}$$

where ρ_{clinical} is the risk on first infection, γ_{clinical} is the number of infections at half-maximum-risk, i.e., when $n = \gamma_{\text{clinical}}$, risk is $\frac{\rho_{\text{clinical}}}{2}$, and δ_{clinical} is proportional to the decay rate with the number of infections at $n = \gamma_{\text{clinical}}$.

There are identifiability problems between the estimates of most of the model parameters due to our broad priors and the nature of fitting a nonlinear model to the age patterns (Notebook S5). We therefore used our very precise estimate for the probability of a blood infection becoming clinical (0.789 ± 0.014) from our analysis of data from Gonçalves et al. (2014) as a very informative prior and refitted the above model. (This parameter is not equivalent to ρ_{clinical} because it is an average over two years of infections, however, due to the slow development of clinical immunity it is a reasonable alternative.)

This fits of the model to the observed annual clinical episode rates are shown in Figure 5.

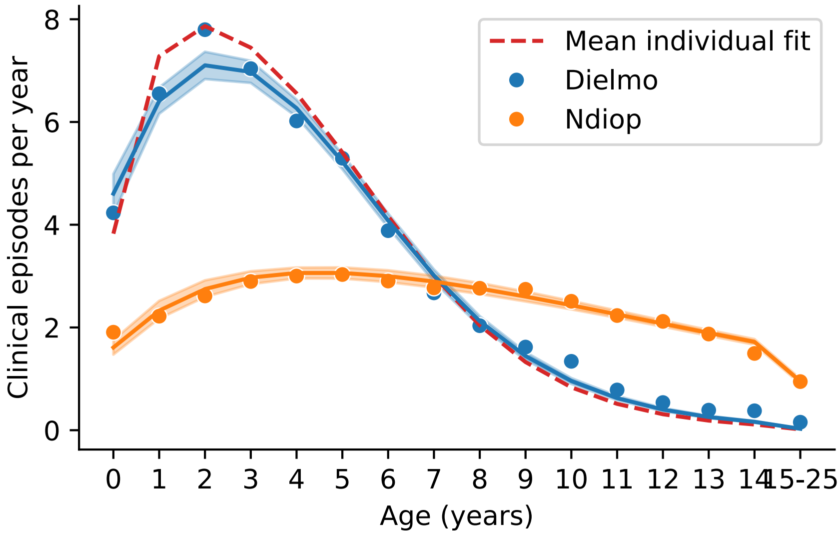

Figure 5: Fit of the clinical episode rate model to age-stratified data from Dielmo and Ndiop villages. Circles are observed annual clinical episode rates. Posterior mean age patterns and their 95% credible intervals are represented by the lines and shaded regions. The red dashed line is the prediction using the mean parameter estimates from the fit to the individual-level data of Dielmo village.

The estimated decay curve is shown in Figure 6. The parameter estimates used in the model are $\rho_{\text{clinical}} = 0.79 \pm 0.02$, $\gamma_{\text{clinical}} = 56.9 \pm 2.1$ infections and $\delta_{\text{clinical}} = 0.033 \pm 0.004$ per infection.

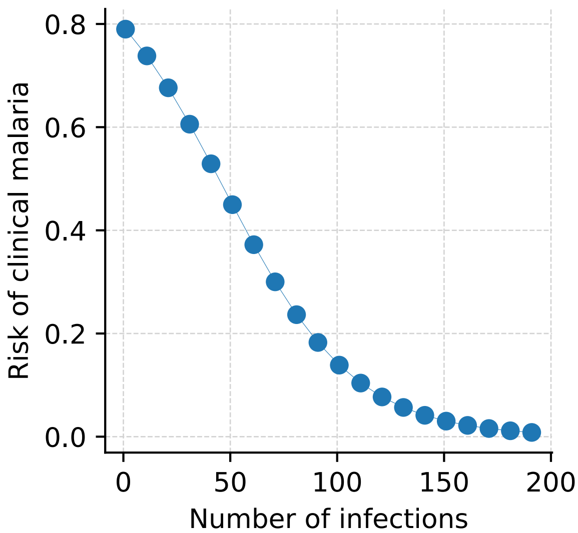

Figure 6: Estimated decline in risk of a blood infection becoming clinical.

7.2 Naturally acquired immunity to severe malaria

Although it is accepted that immunity against severe malaria develops more rapidly than immunity against clinical disease, there is still controversy about how rapidly it develops. Baird et al. (1998) has shown that non-immune adults develop immunity against delirium or loss of consciousness (measured by the number of medical evacuations) within 4 months of migrating to a hyperendemic malaria region. Gupta et al. (1999) estimated immunity developing after one or two infections based on model fitting to incidence of severe disease in hospital admissions in The Gambia and Kenya across a range of transmission intensities. Muñoz

Sandoval et al. (2025) analysed data from Gonçalves et al. (2014) to demonstrate that the risk of severe malaria decays at a rate of 0.12 per blood infection: a half-life of 5.7 infections. In contrast, Griffin et al. (2015) suggest that the risk of severe malaria depends on EIR: At low EIR risk declines rapidly with exposure, but at high EIR risk rises slowly then declines slowly with exposure over a few decades.

Our initial model of the decline in risk of severe malaria with exposure is

$$\Pr(\text{clinical episode becomes severe on } n^{\text{th}} \text{ infection at age } a) = \rho_{\text{severe}} e^{-\delta_{\text{severe}}[n-1]}$$

where ρ_{severe} is the risk on first infection and δ_{severe} is the rate of decline in risk with exposure. We re-estimated these parameters using the data from Gonçalves et al. (2014). They recorded all infections and severe malaria episodes in a cohort of 882 children from birth till two to four years old in Tanzania from 2001-2004. Our analysis in Notebook S4 is a more comprehensive analysis of the data than was performed in Muñoz Sandoval et al. (2025), although the results are very similar. We estimated the average risk on first infection to be $\rho_{\text{severe}} = 4.2\% \pm 0.5\%$. We estimated the average decay in risk per infection to be $\delta_{\text{severe}} = 0.12 \pm 0.03$ - which corresponds to a halving of risk every 5.7 infections. However, this half-life might be an overestimate due to amodiaquine treatment failure causing recrudescent infections to be counted as independent infections. Gonçalves et al. (2014) reported amodiaquine 28-day failure rate of at least 52%. Another study put the 28-day failure rate in Tanzania in 2005 at 76% (Mutabingwa et al. 2005). We examine this uncertainty in Notebook S10.

However, as discussed in Notebook S6, this model of severe malaria risk is not entirely consistent with other severe malaria and mortality age-pattern datasets. In particular, at low transmission intensities severe episode rate and mortality rate remain high into the early teens causing a second peak in deaths (Figure 7) due to rising death rate with age after age four (Reyburn et al. (2005); Notebook S7). This causes long-term cumulative deaths to rise with declining transmission intensity: a pattern not seen in the epidemiological data (Abdullah et al. 2007).

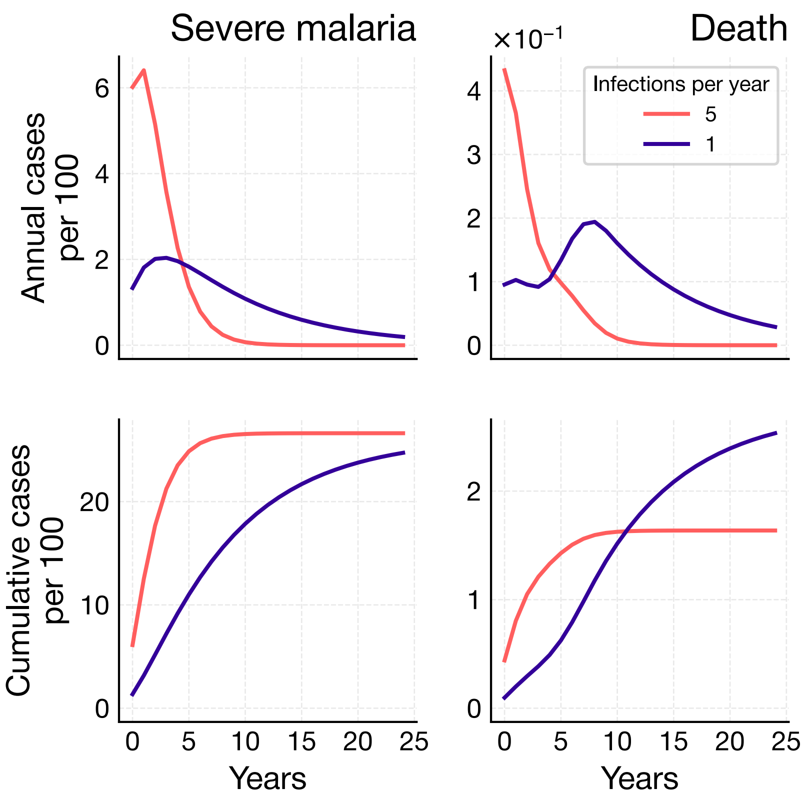

Figure 7: Severe malaria and mortality rates and cumulative episodes for 1 and 5 infections per child per year assuming exposure-only dependent reduction in risk of severe malaria.

To overcome this discrepancy we propose a modified model with risk reducing with age-dependent exposure:

$$\Pr(\text{clinical episode becomes severe on } n^{\text{th}} \text{ infection at age } a) = \rho_{\text{severe}} e^{-[\delta_{\text{severe}} + \epsilon_{\text{severe}} a][n-1]}$$

where ϵ_{severe} modifies the reduction in risk of severe malaria with age a. In Notebook S6 we show that a value of $\epsilon_{\text{severe}} = 0.050$ per infection per year is sufficient to suppress the second peak in deaths and makes long-term cumulative deaths fall with falling transmission intensity (Figure 8) as is reported in the literature (Abdullah et al. 2007).

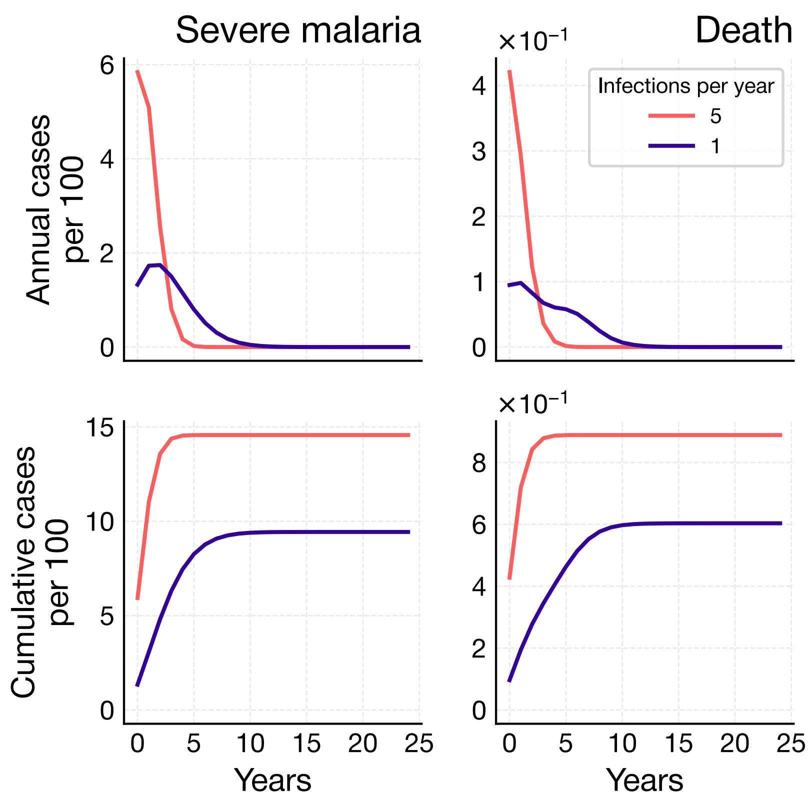

Figure 8: Severe malaria and mortality rates and cumulative episodes for 1 and 5 infections per child per year with the modified model of severe malaria risk reducing with age-dependent exposure.

The resulting risk curves are shown in Figure 9. Irrespective of age, the risk of a clinical episode on first infection becoming severe is 4.2%. However, older children with the same number of infections as younger children are less at risk of severe malaria.

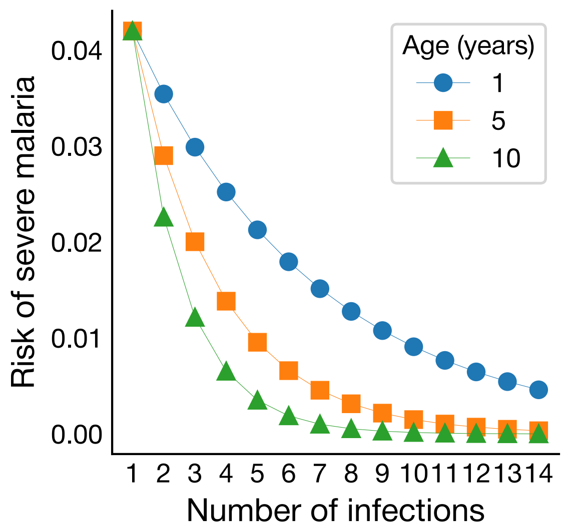

Figure 9: Risk of a clinical episode becoming severe depends on age and exposure.

8 Other model assumptions

Model assumptions that differ from the WHO consensus models. We do not model:

1. super-infections,
2. a delay from infection to occurrence of symptoms and death as this has no impact on long-term dynamics,
3. within-host parasite density,
4. duration of infection, therefore we do not predict prevalence, only incidence.

9 Definition of averted cases and vaccine efficacy

In malaria vaccine clinical trials, and modelling studies of those trials, vaccine efficacy is usually calculated from the last primary dose onwards. We take the same approach here because of the lack of data and the large uncertainty in estimating partial protection before the last primary dose. In other words, we do not model partial protection before the last primary dose and we start counting infections, clinical and severe malaria episodes and deaths in the unvaccinated and vaccinated cohorts immediately after the last primary dose.

The time-dependent number of averted cases is therefore

$$\text{cases averted } (t) = \text{cumulative unvaccinated cases } (t) - \text{cumulative vaccinated cases } (t)$$

where $t = 0$ is the day of the last primary dose in the vaccinated cohort. This equation applies to infections, clinical and severe malaria episodes and deaths.

Time-dependent efficacy is the number of averted cases divided by the cumulative unvaccinated cases after time t:

$$\text{efficacy } (t) = \frac{\text{averted cases } (t)}{\text{cumulative unvaccinated cases } (t)}$$

The interpretation of this definition of efficacy is the proportional reduction in the number of infections, or a clinical or severe malaria episodes or deaths in the vaccinated cohort compared to the unvaccinated cohort **given that everyone in the vaccinated cohort is fully vaccinated**. Therefore, it ignores any infections, clinical and severe malaria episodes or deaths before everyone in the cohort is fully vaccinated.

Using this definition, averted cases can be used to directly compare the effectiveness of vaccines or vaccine protocols. The above definition of efficacy, however, cannot because of possible differences in the timing of a vaccine's last primary dose. For example, R21 is given at months 0, 1 and 2. Whereas RH5.1 is given at months 0, 1, and 5. Thus the denominator in the efficacy equation is calculated over a longer period of time for R21 than RH5.1.

To mitigate this problem with efficacy, and to allow us to use efficacy to compare vaccines, for RH5.1 we start counting cases at the same time as we would start counting cases for R21 (i.e., the last primary dose of R21). For the combined R21+RH5.1 vaccine we also start counting cases at the last primary dose of R21.

10 Pre-erythrocytic vaccines (RTS,S and R21)

Figure 10 illustrates the possible effects of a pre-erythrocytic on infection. It is not known if a pre-erythrocytic vaccine prevents infection of the liver, or clears a liver infection, or both. But it does reduce the risk of an infectious bite resulting in a blood-stage infection.

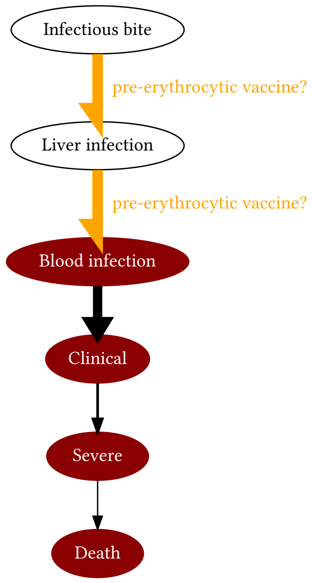

Figure 10: The possible effects of a pre-erythrocytic vaccine.

We make the following assumptions about pre-erythrocytic vaccines:

- they reduce the risk of a blood-stage infection
- they do not directly affect the rate of acquisition of natural immunity with exposure, but they do so indirectly by preventing a blood-stage infection
- initial efficacy and decay rate are not affected by any other characteristics such as age, previous exposure, sex, etc.
- the narrow immunisation age-range of 5-17 months means the vaccine has minimal impact on overall transmission or prevalence (Galactionova, Smith, and Penny 2021) which obviates the need to model transmission.

We define vaccine protection of a pre-erythrocytic vaccine as "the probability that vaccine-induced immune responses prevent a blood infection". This has been estimated for the RTS,S vaccine for 3-dose and 3-dose + booster schedules (White et al. 2015) and for the 3-dose + booster for R21 (Schmit et al. 2024). We adopt the Imperial vaccine profile estimates because they have estimates for both RTS,S and R21 vaccine profiles using the same methodology (Figure 11).

Schmit et al. (2024) may have slightly over-estimated the protective effect of R21. The clinical trial on which their estimates were made used SMC. But the model Schmit et al. (2024) used to fit to the trial data did not include SMC (see model limitations on page 473). So their estimated protective effect of R21 also includes the protective effect of SMC. Our informal fitting to the trial data suggests a 5% over-estimation. This has a negilible effect on model outputs.

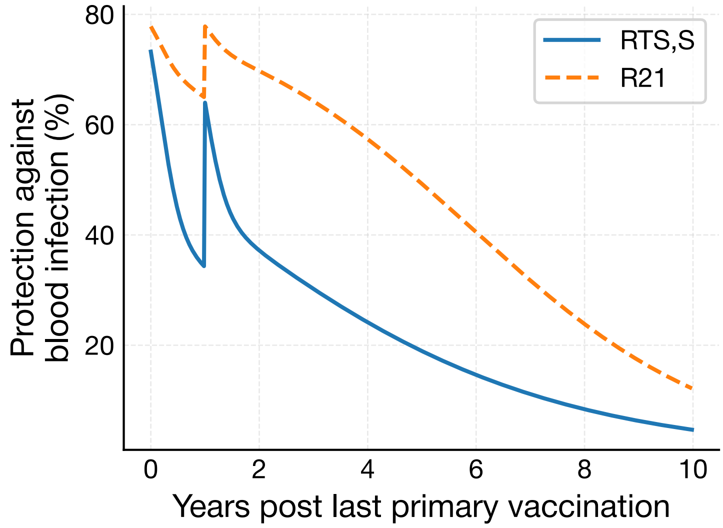

Figure 11: Imperial's estimated best fits of RTS,S and R21 vaccine protection profiles after the third dose of the primary course, as estimated from phase 3 data in children receiving their first dose between 5 and 17 months and a booster 12 months after the third dose. The R21 curve has been adjusted for possible bias.

11 Blood-stage vaccine (RH5.1)

Figure 12 illustrates the possible effects of a blood-stage vaccine on infection and disease. The RH5.1 vaccine blocks merozoite invasion of erythrocytes thereby reducing parasite growth rate and parasitaemia at presentation. This may cause any of the following:

- reduced risk of a blood infection (similar to a pre-erythrocytic vaccine)
- reduced risk of a blood infection causing clinical symptoms
- reduced risk of clinical symptoms becoming severe

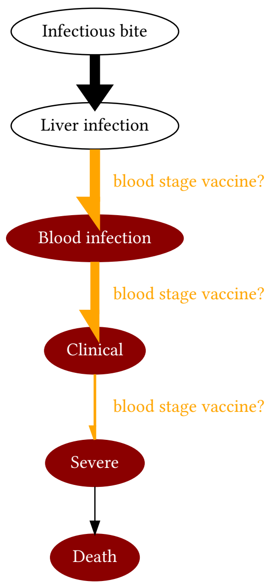

Figure 12: The possible effects of a blood-stage vaccine.

We were kindly given unpublished data from the phase 2b trial of the RH5.1/Matrix-M blood stage vaccine (Natama et al. 2024). The vaccine was given to children aged 5–17 months in Nanoro, Burkina Faso. The trial had three arms: the control "rabies" arm with the rabies vaccine Rabivax-S, the "delayed" vaccine arm with regimen of RH5.1/Matrix-M at 0, 1, and 5 months and the "monthly" vaccine arm with regimen at 0, 1, and 2 months. Vaccinations were completed part way through the malaria season.

As we discuss in Notebook S9 we are not able to separately quantify each of the above three reduction in risks because none of the animal or human vaccine data allows us to determine whether RH5.1 reduces the risk of blood infection or not.

Instead, we examine two extreme scenarios: one where RH5.1 reduces the risk of blood infection but not clinical malaria, and the other where RH5.1 reduces the risk of clinical malaria but not blood infection. The reality is probably somewhere in between. But by examining these two extremes we can robustly test our conclusions against this uncertainty.

The phase 2b trial dataset contains cumulative all-cause fevers for each arm (Figure 13A) and cumulative fevers with detectable parasites by microscopy (>0 parasites/μl with fever in the last 24 h; Figure 13B). We only use the delayed vaccine arm in our model as the efficacy of the monthly vaccine arm is inferior. Using this data with some reasonable assumptions we calculate that, for the first scenario, RH5.1 reduces the risk of blood infection by 43%, and for the second scenario RH5.1 reduces the risk of a blood infection becoming clinical by 50% (Notebook S9).

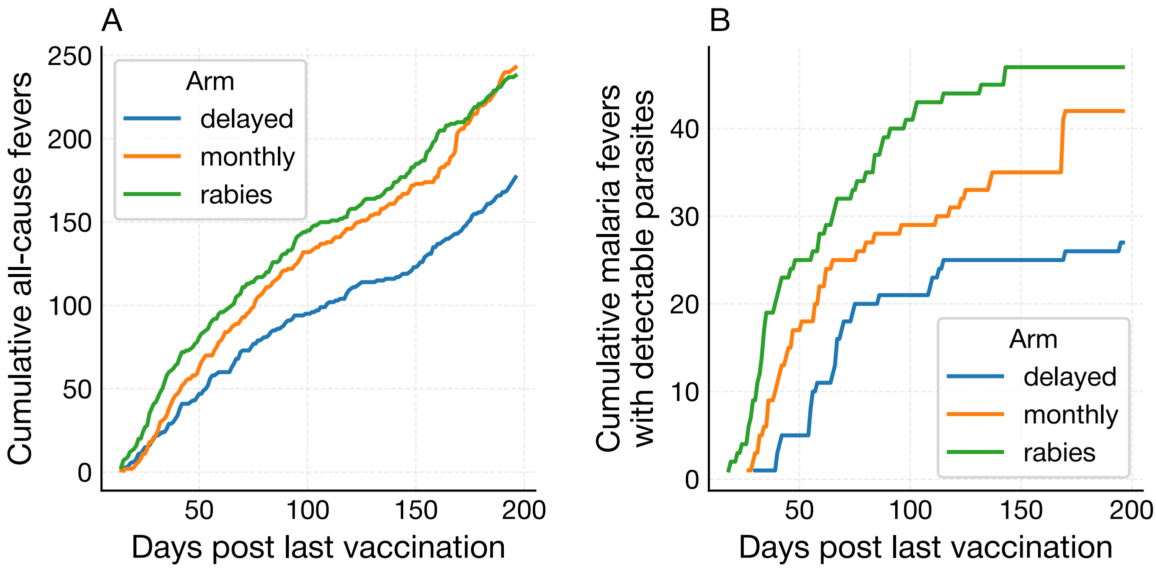

Figure 13: A) Cumualtive all-cause fevers recorded either actively or passively for 200 days post last primary vaccination in the three trial arms of the phase 2b trial of the RH5.1/Matrix-M blood stage vaccine (Natama et al. 2024). B) Cumulative clinical episodes (>0 parasites/μl by microscopy with fever in the last 24 h) in the three arms.

Although there is no trial data on severe malaria episodes we can estimate the reduction in risk of severe malaria using parasitaemia data from the phase 2b trial (Figure 14). The delayed arm caused a 3.2 fold reduction in parasitaemia at presentation. Using the relationship between parasitaemia and the risk of severe malaria that we calculated from the data in Gonçalves et al. (2014) (Notebook S4) we estimate that the delayed arm reduced the risk of severe malaria by 29% (see Notebook S9).

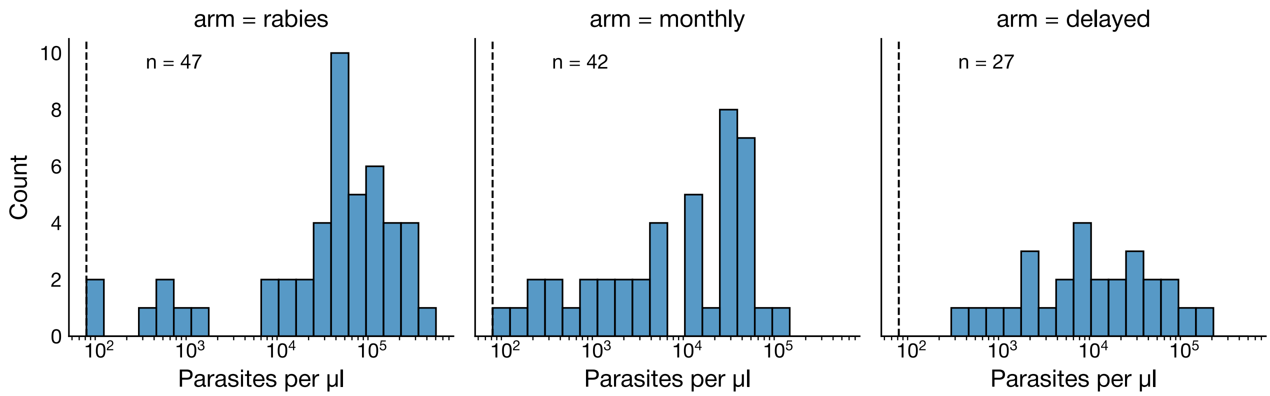

Figure 14: Distributions of parasitaemia at presentation in each trial arm. The vertical dashed line in each panel is the smallest parasitaemia of 72 parasites/µl recorded across all episodes.

These reduction in risks are averaged over a single malaria season. However, the protective effect of RH5.1 probably wanes rapidly over about six months based on anti-RH5.1 IgG antibody profiles (Silk et al. 2024) (Figure 15). We use these profiles to model the waning efficacy of RH5.1 (see Notebook S9).

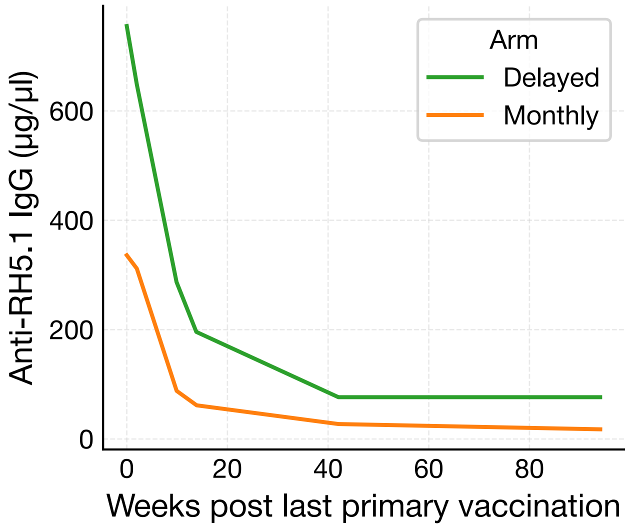

Figure 15: Median anti-RH5.1 IgG antibody profiles as measured by ELISA of the delayed and monthly arms reproduced from Figure S4 of Silk et al. (2024). The profiles are from the 10D group of children two weeks after the last primary dose.

12 Seasonal malaria chemoprevention

Seasonal malaria chemoprevention (SMC) is widely used in areas with highly seasonal transmission. It is given to young children monthly, four to five times during the peak season. SMC typically uses a combination of two drugs, sulfadoxine–pyrimethamine and amodiaquine. Amodiaquine has a relatively short half-life and is intended to quickly clear any existing malaria parasites in a child's system. Sulfadoxine–pyrimethamine has a longer half-life to prevent new infections for about 28 days (Zongo and Compaoré 2024).

To parameterise SMC for our model we made use of five year phase 3 trial data from Chandramohan et al. (2021) and Dicko et al. (2024). See Notebook S8 for more details. This was a trial of SMC-only, RTS,S-only and SMC+RTS,S combined, in two seasonal sites: Bougouni district in Mali and Hounde district, Burkina Faso.

We tried fitting several different protection profiles for SMC against blood infection. We finally settled on one in which protection ramps up from the start of the season at some given rate, until it plateaus at a given coverage. It stays at this level until the end of four rounds (four months) at which point it returns to zero (Figure 16).

This model gave excellent fits to the trial data (Notebook S8).

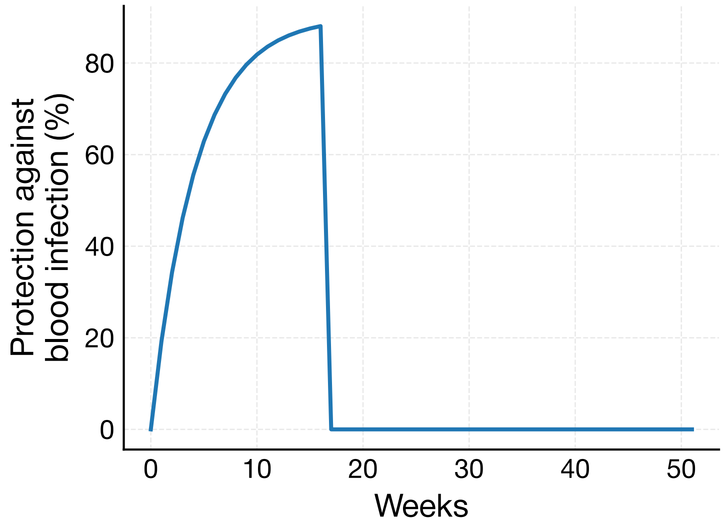

Figure 16: Protection profile of SMC calibrated to five year phase 3 trial data from Chandramohan et al. (2021) and Dicko et al. (2024). Protection rises exponentially to a maximum of 90% coverage, then ceases after four months.

13 Mathematical description of the model

Let $\mu(a)$ be the natural non-malaria mortality rate of people of age a weeks for $a = 0, 1, 2...$ (This Notebook).

Let $E_{\text{infection}}(a)$ be the relative early life protection against blood-stage malaria infection for children of age a (Notebook S3).

Let $L_{\text{infection}}(t)$ be the protective efficacy of a pre-erythrocytic vaccine against blood infection in week t post last primary dose, where $t = 0, 1, 2, ...$ (This Notebook). Let $B_{\text{infection}}(t)$ be the protective efficacy of a blood-stage vaccine against blood infection in week t (Notebook S9). Let $F_{\text{infection}}(t)$ be the protective efficacy of SMC against blood infection in week t (Notebook S8).

Let λ be the time-averaged annual blood infection rate. Let $s(t)$ be the seasonal modification to this rate such that $\sum_{t=0}^{51} s(t) = 1$ and $s(t) = s(t (\text{mod } 52))$ (Notebook S3).

Let b be a random variable that represents relative bite rate/susceptibility to infection. The expected value of b is 1.

Therefore, the force of infection experienced by people of age a with bite rate/susceptibility b, fully vaccinated t weeks ago is

$$\Lambda(b, a, t) = \frac{b\lambda s(t)}{52}[1 - L_{\text{infection}}(t)][1 - B_{\text{infection}}(t)][1 - F_{\text{infection}}(t)][1 - E_{\text{infection}}(a)]$$

At $t = 0$ children between ages $a_{\min}$ and $a_{\max}$ weeks receive their last primary vaccine dose. We assume that people can have at most 1 infection per week. Therefore, people at age a have from 0 to a infections.

Therefore at a given time t, the youngest people will have age $a_{\min} + t$ and between 0 and $a_{\min} + t$ infections. The oldest people will have age $a_{\max} + t$ and between 0 and $a_{\max} + t$ infections.

Let $N(n - 1, b, a - 1, t - 1)$ be the number of people of age $a - 1$ weeks with $n - 1$ infections and bite rate/ susceptibility b in week $t - 1$.

In the next week, t, a proportion $\mu(a)$, will die of non-malaria causes and the surviving will experience a force of infection of $\Lambda(b, a, t)$. Then the number of new blood infections in these people, $I(n, b, a, t)$ is

$$I(n, b, a, t) = [1 - \mu(a)]\Lambda(b, a, t)N(n - 1, b, a - 1, t - 1)$$

Let $R_{\text{clinical}}(n)$ be the risk of clinical malaria in people with n infections (Notebook S5). Let $B_{\text{clinical}}(t)$ be the protective efficacy of a blood-stage vaccine against a blood infection becoming clinical in week t (Notebook S9). Therefore the number of new clinical episodes in these people is

$$C(n, b, a, t) = [1 - B_{\text{clinical}}(t)]R_{\text{clinical}}(n)I(n, b, a, t)$$

Let $R_{\text{severe}}(n, a)$ be the risk of clinical malaria becoming severe in people of age a with n infections (Notebook S4; Notebook S6). Let $B_{\text{severe}}(t)$ be the protective efficacy of a blood-stage vaccine against a clinical episode becoming severe in week t (Notebook S9). Therefore the number of clinical episodes that become severe in these people is

$$S(n, b, a, t) = [1 - B_{\text{severe}}(t)]R_{\text{severe}}(n, a)C(n, b, a, t)$$

Let $R_{\text{death}}(a)$ be the risk of severe malaria-associated death of people of age a (Notebook S7). Therefore the number of severe malaria episodes that result in death in these people is

$$D(n, b, a, t) = R_{\text{death}}(a)S(n, b, a, t)$$

Model output is, for each week, number of new infections, number of clinical episodes that do not become severe, number of severe episodes and number of severe episodes that result in death.

Finally we update the number of people of age $a = a_{\min} + t, ..., a_{\max} + t$ weeks experiencing bite rate b in week $t > 0$:

$$N(0, b, a, t) = [1 - \mu(a)]N(0, b, a - 1, t - 1) - I(1, b, a, t) \text{ for } \quad n = 0$$
$$N(n, b, a, t) = [1 - \mu(a)]N(n, b, a - 1, t - 1) - I(n + 1, b, a, t) + I(n, b, a, t) - D(n, b, a, t) \text{ for } \quad n = 1..., a - 1$$
$$N(a, b, a, t) = I(n, b, a, t) - D(n, b, a, t) \text{ for } \quad n = a$$

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